# VEINS & LYMPHATIC OF HEAD & NECK

# ***LEARNING OBJECTIVES***

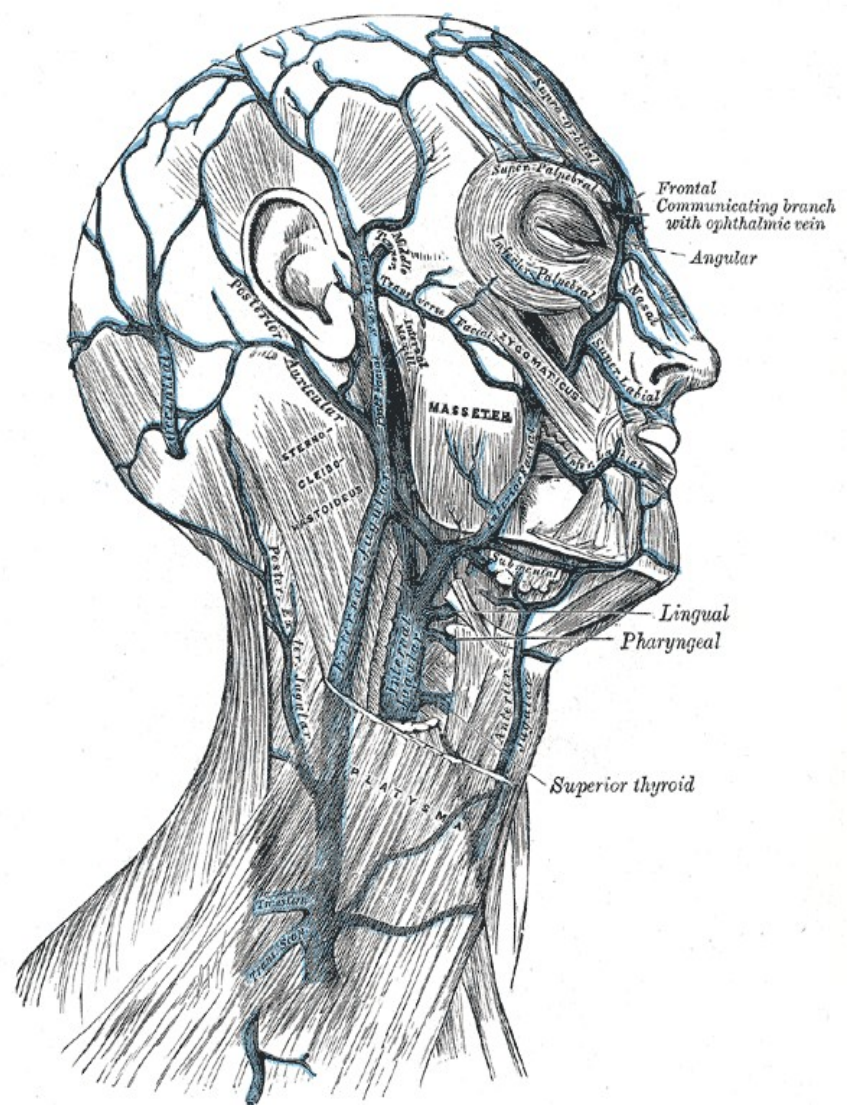
- At the end of lecture student should be able to:
  - Describe venous drainage of head, face & neck.
  - Lymph nodes of head & neck, their location & area of drainage.

# ***THE VEINS OF THE HEAD AND NECK***

- The veins of the head and neck may be subdivided into three groups:
  1. The **veins of the exterior of the head and face.**
  2. The **veins of the neck.**
  3. The **diploic veins,** the **veins of the brain,** and the **venous sinuses of the dura mater.**

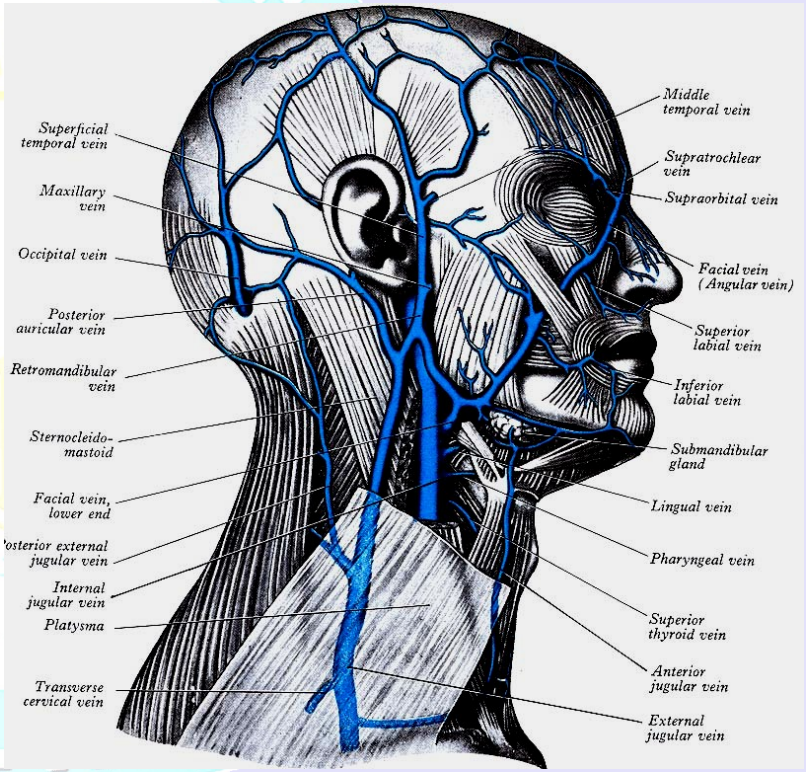
# THE VEINS OF THE EXTERIOR OF THE HEAD AND FACE

- The veins of the exterior of the head and face are:
  1. Frontal
  2. Superficial Temporal
  3. Supraorbital
  4. Internal Maxillary
  5. Angular
  6. Posterior Facial
  7. Anterior Facial
  8. Posterior Auricular
  9. Occipital.



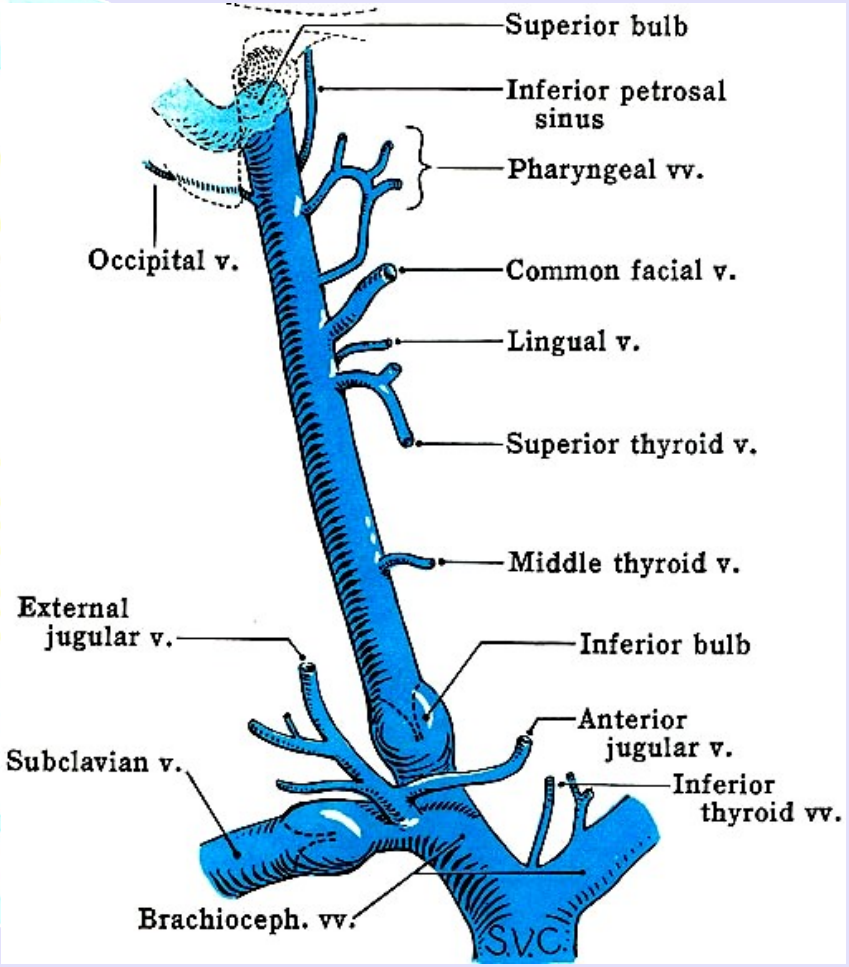
# VEINS OF THE HEAD AND NECK

- Venous drainage from the face is entirely superficial
- All the venous drainage from the head and neck terminate in the *internal jugular vein* which join the *subclavian vein* to form the brachiocephalic vein behind the medial end of the clavicle



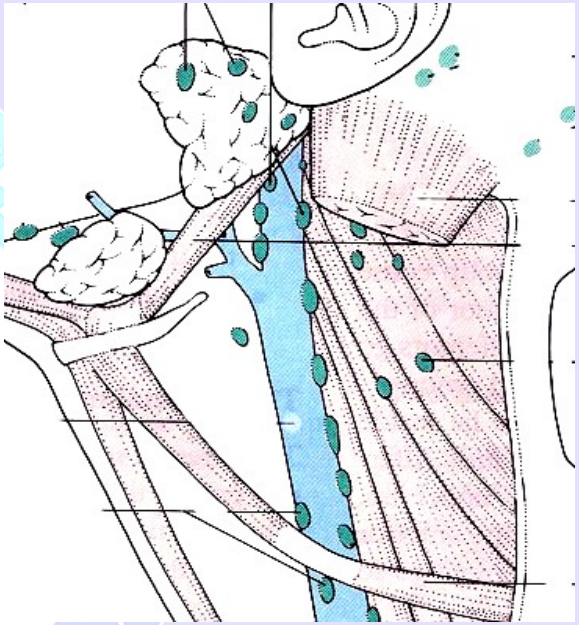
# VEINS OF THE HEAD AND NECK

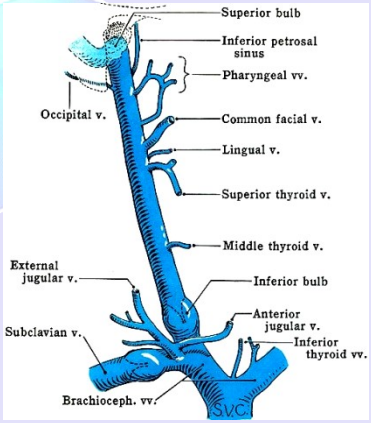
- ***INTERNAL JUGULAR VEIN:***
  - **It receive blood from the brain, face and the neck.**
  - **It emerges through the jugular foramen,as a continuation of the sigmoid sinus descend down in the neck, first behind then lateral to the internal carotid artery inside the carotid sheath**



# VEINS OF THE HEAD AND NECK

- The deep cervical lymph nodes are adjacent to the vein
- Terminate beneath the triangular interval between the sternal and the clavicular head of the *sternocleidomastoid* muscle joining the subclavian vein to form the *brachiocephalic* vein

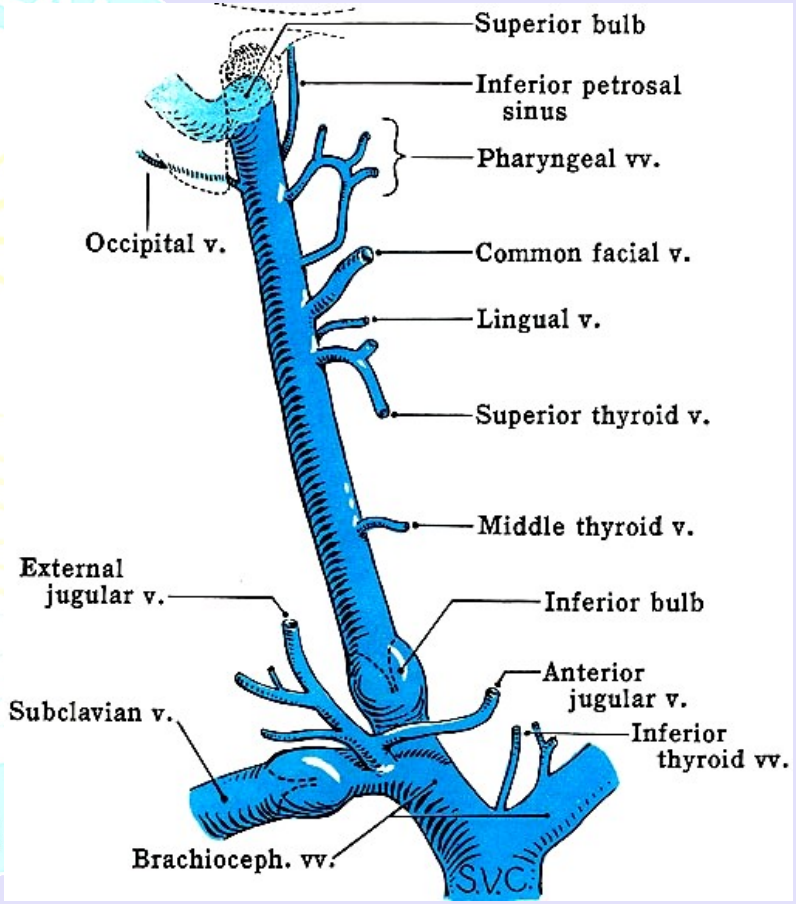




# ***VEINS OF THE HEAD AND NECK***

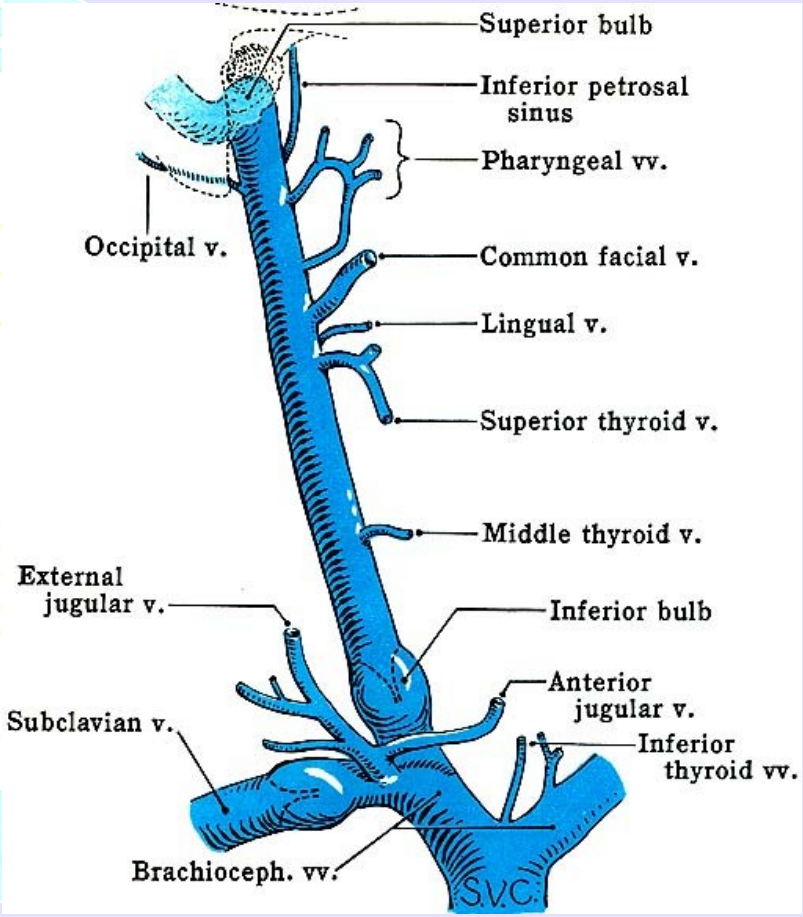
- The superior and inferior bulbs:
  - are dilatation near the origin and the termination of the vein.
  - Above the inferior bulb is a bicuspid valve

- Skin, superficial fascia, platysma, deep cervical fascia and parotid lies lateral and anteriorly



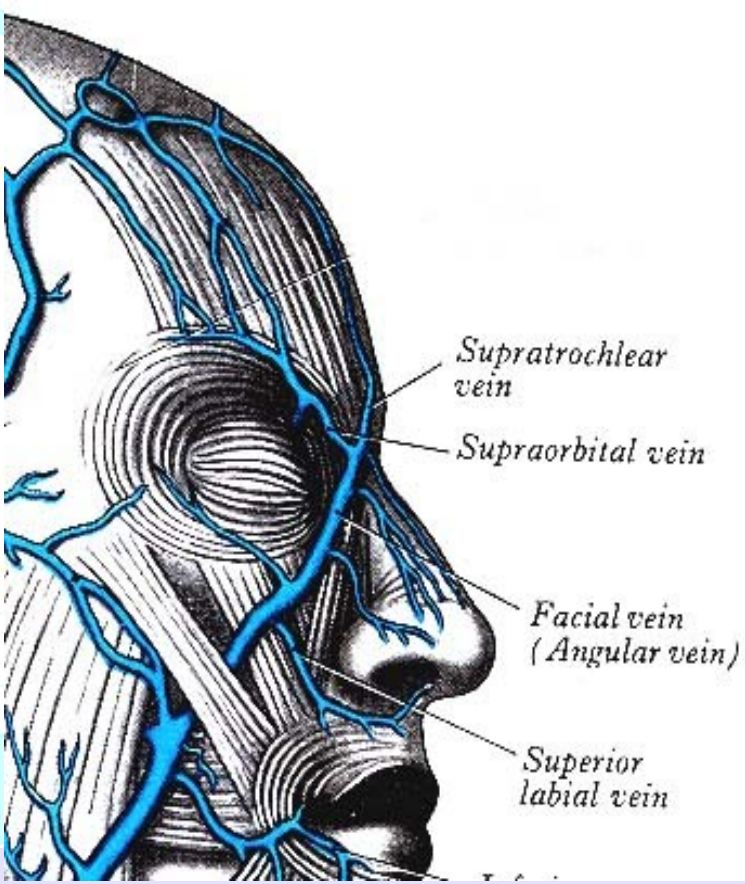
# ***VEINS OF THE HEAD AND NECK***

- Tributaries:
  - ***Inferior petrosal sinus:***
    - **assist in draining the cavernous sinus**
    - **leaves through anterior part of jugular foramen**
    - **join the vein below the superior bulb**



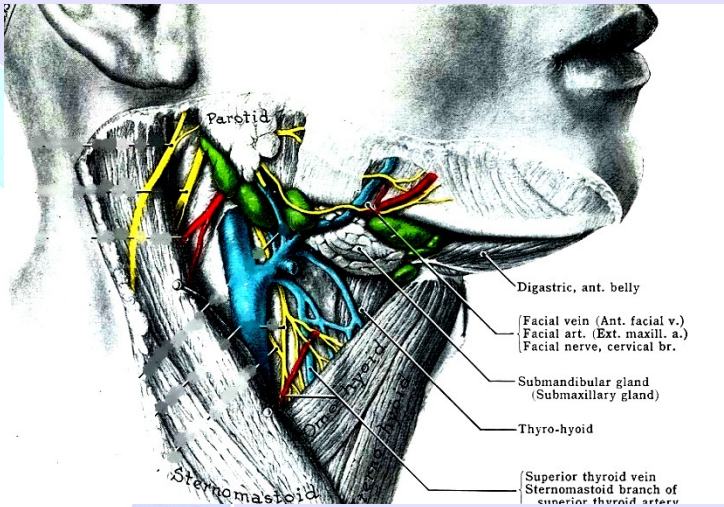
# VEINS OF THE HEAD AND NECK

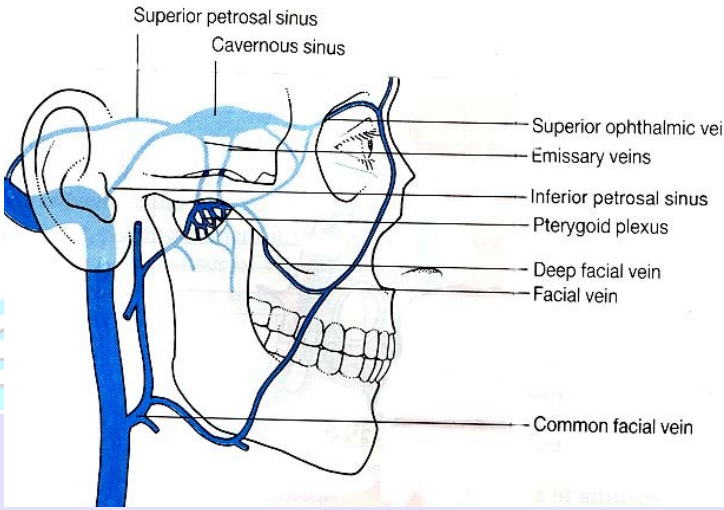
- ***Facial vein:***
  - **is formed by the union of the supraorbital and supratrochlear veins the medial canthus to form the angular vein**
  - **Communicate with the cavernous sinus through the ophthalmic vein via the supraorbital**



# VEINS OF THE HEAD AND NECK

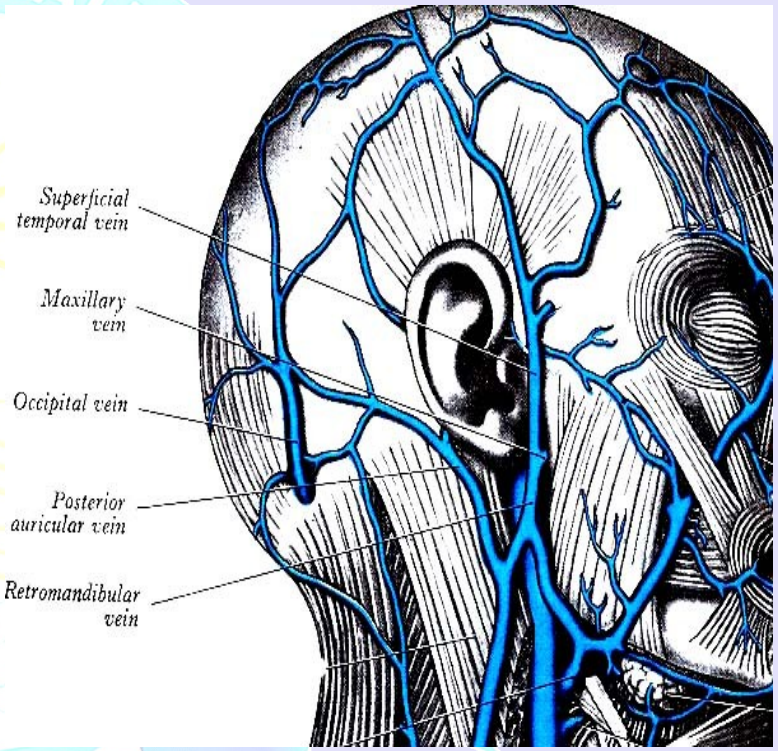
- ***FACIAL VEIN:***
- **descend on the face behind the facial artery to the lower border of the mandible**
- **to be joined by the anterior division of the *retomandibular vein***
- **Joins the:**
  - **pterygoid plexus through *deep facial vein***
  - **Cavernous sinus through *superior ophthalmic vein***



# VEINS OF THE HEAD AND NECK

- ***Retromandibular vein:***
  - **formed by the union of *superficial temporal* and *maxillary vein* from the pterygoid plexus**
  - **passes downwards in the substance of the parotid gland emerging from its lower border & divide into two divisions**

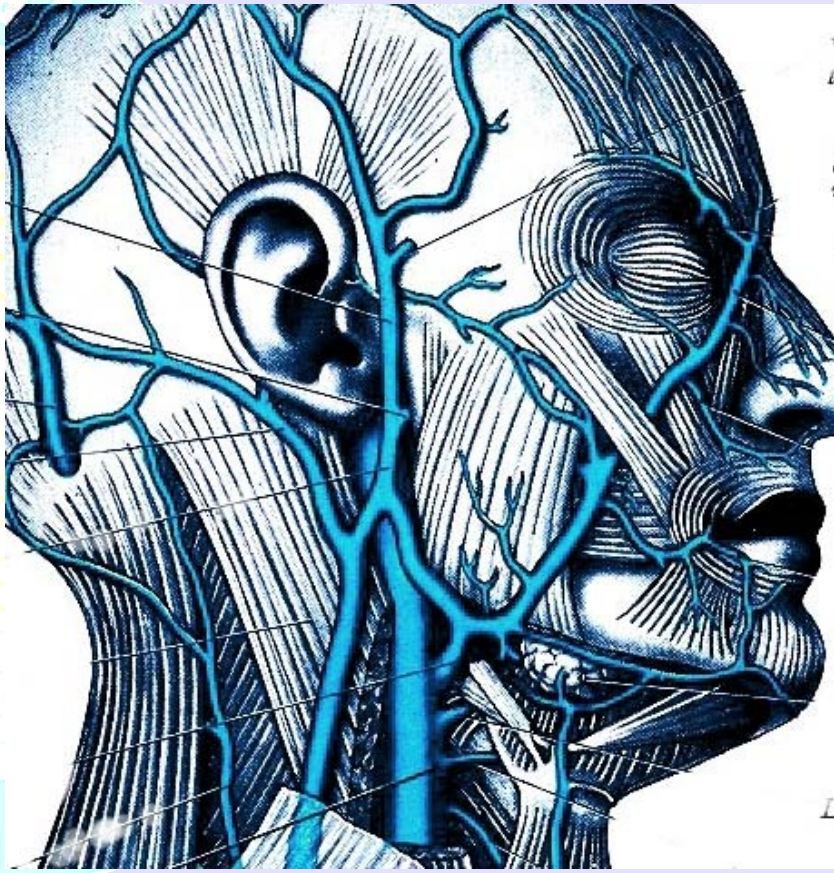


# ***VEINS OF THE HEAD AND NECK***

- ***Retromandibular vein:***
- Anterior division:
  - joins the facial vein

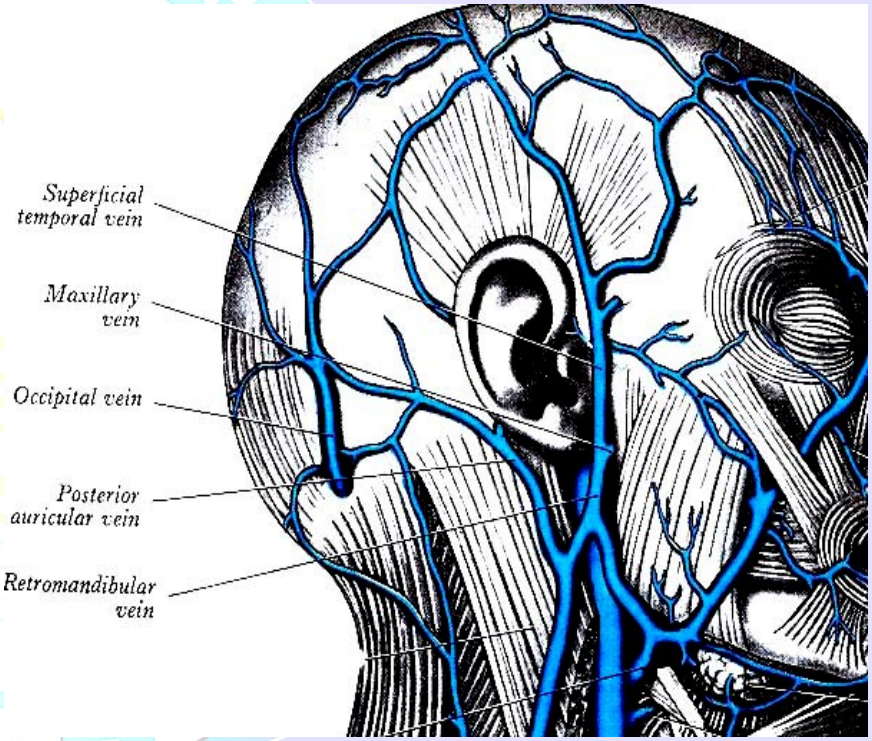
- Posterior division:
  - pierces the deep fascia and join the ***posterior auricular*** to form the ***external jugular.***

- It empty into the ***subclavian vein***



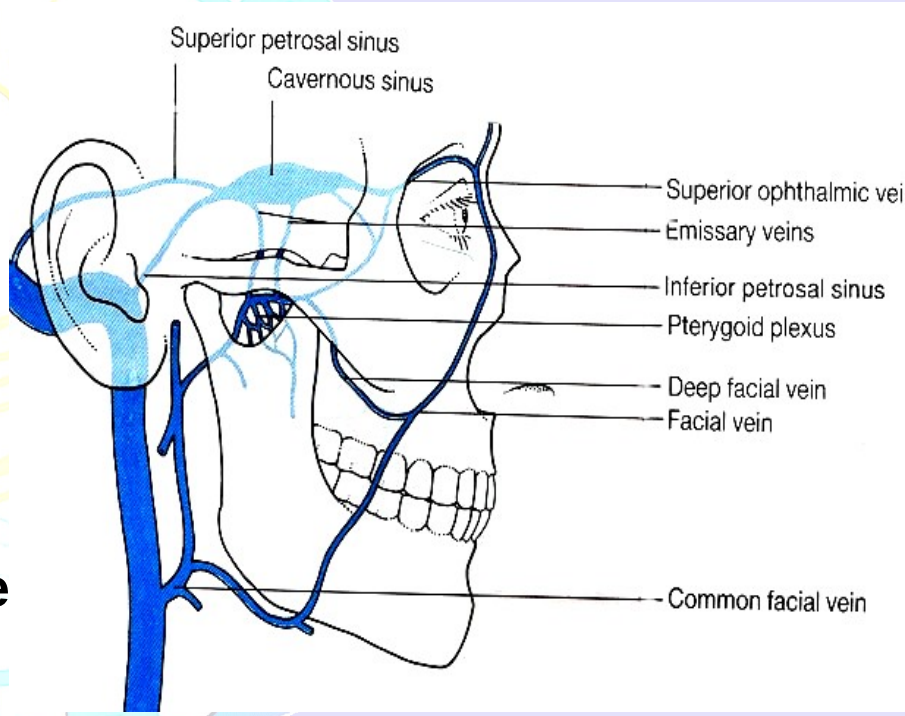
# ***VEINS OF THE HEAD AND NECk***

- ***The maxillary vein:***
  - A short trunk accompany the first part of the artery.
  - Formed by confluence of the veins of the pterygoid plexus.
  - It passes backward between the sphenomandibular ligament and the neck of the mandible
  - Unite with the ***superficial temporal vein*** to form the ***retromadibular vein.***



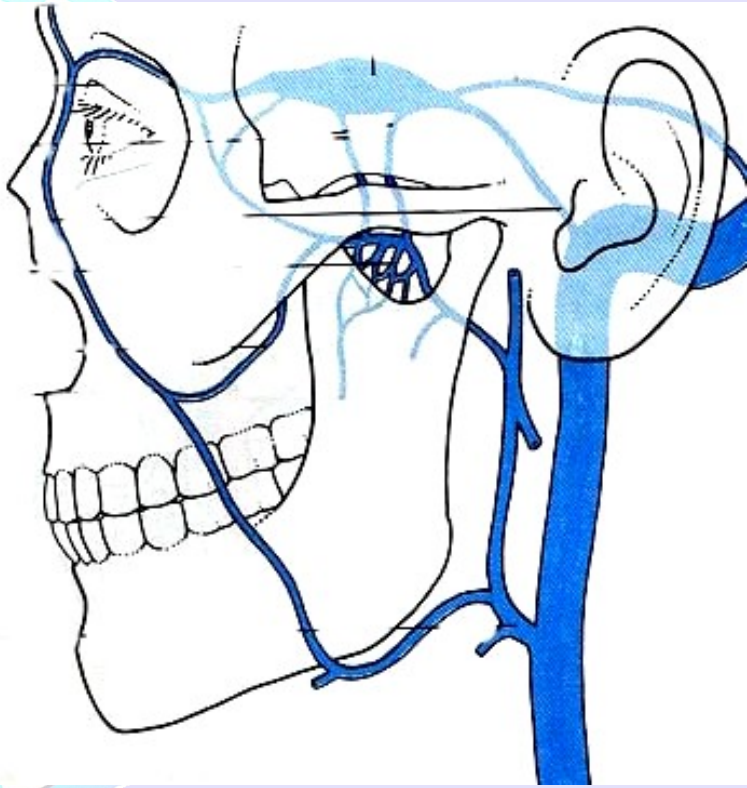
# VEINS OF THE HEAD AND NECK

- ***Pterygoid plexus:***
  - **A network of very small veins, lie around and within the lateral pterygoid muscle in the infratemporal region**
  - **receive some of the veins that correspond to the maxillary artery, inferior ophthalmic vein (internal carotid blood) and the deep facial vein.**



# ***VEINS OF THE HEAD AND NECK***

- ***Pterygoid plexus:***
- Drain into a pair of large, short maxillary veins which join the superficial temporal vein to form the ***retromandibular.***



- Deep facial vein drain the plexus into the ***facial vein*** if the maxillary is occluded

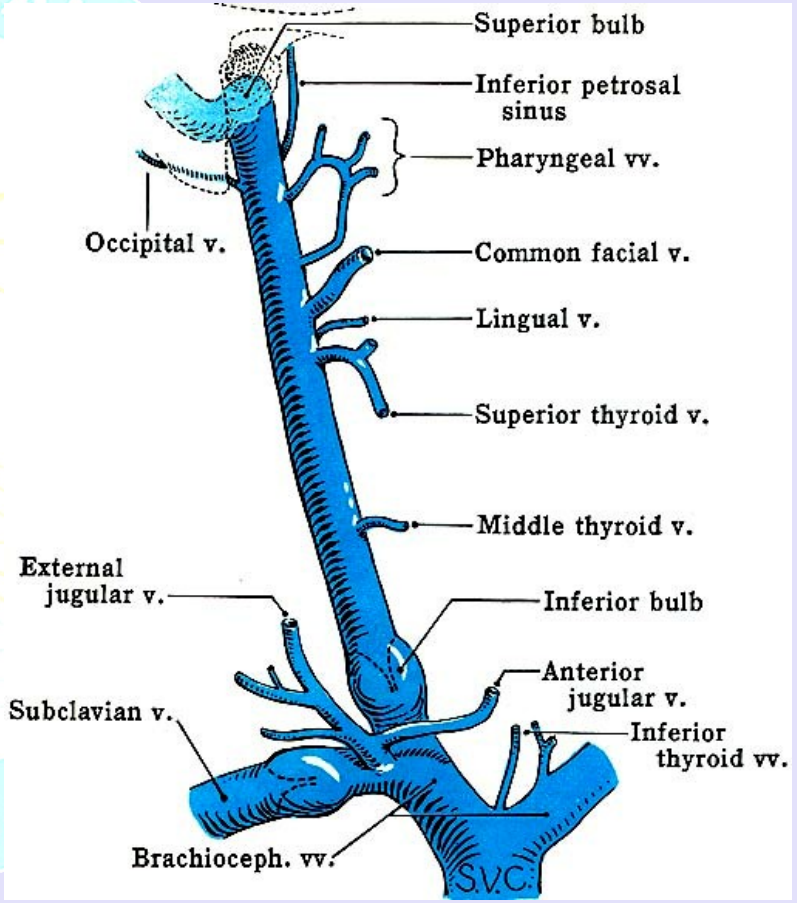
# ***VEINS OF THE HEAD AND NECK***

- ***PTERYGOID PLEXUS:***
- Act as ***peripheral pump***, to aid venous return by the pumping action of the muscle every time the mouth is opened.

- ***Yawing***, a prolonged and forcible contraction of the lateral pterygoid to open the mouth, is accompanied by contraction of the diaphragm and stretching of limbs, is a reflex triggered by venous stagnation

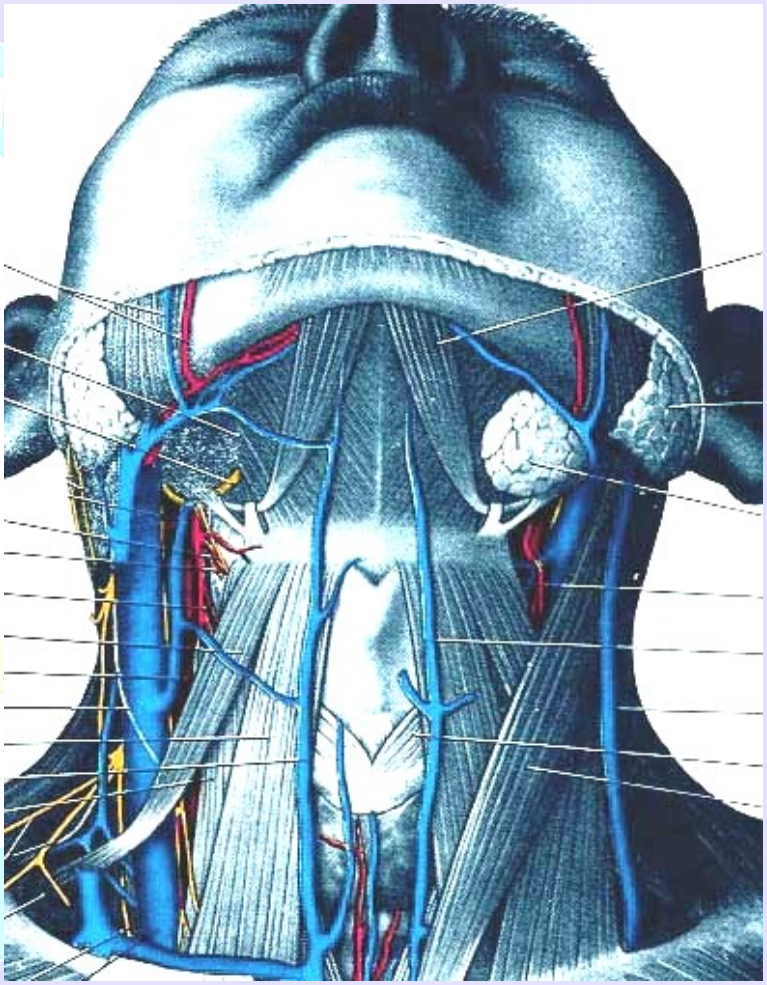
# VEINS OF THE HEAD AND NECK

- ***Pharyngeal veins:***
  - drain the pharyngeal plexus on the outer surface of the pharynx.
  - End in the internal jugular vein , the facial the lingual or the superior thyroid veins



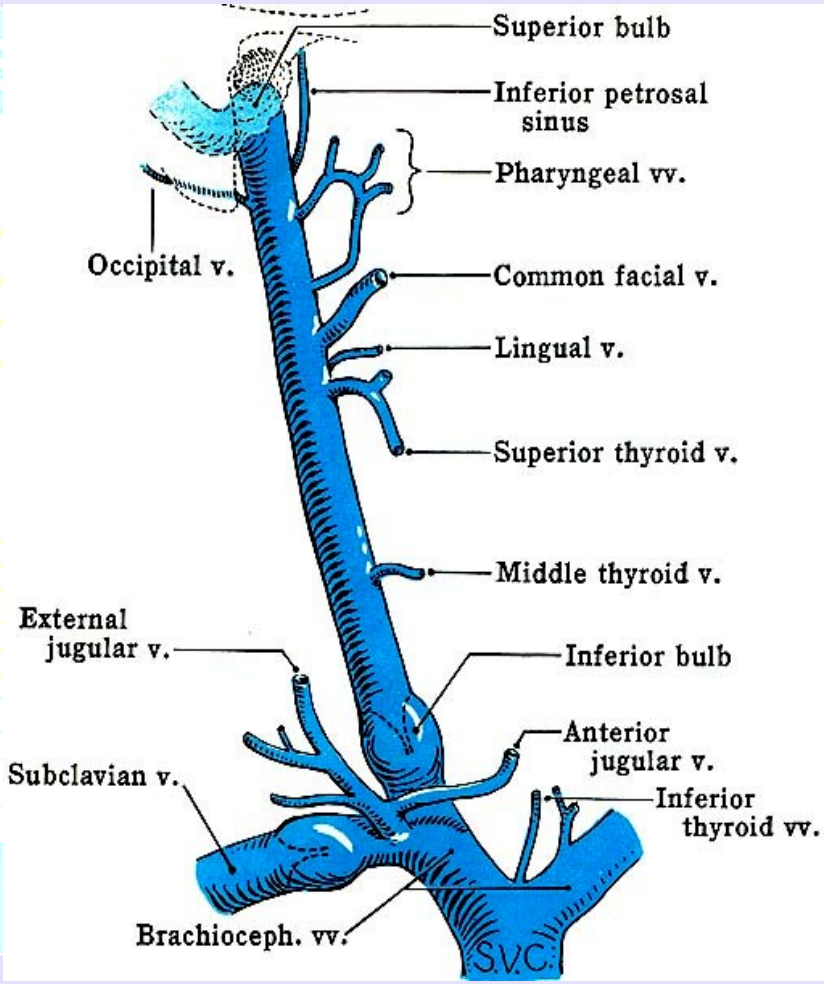
# ***VEINS OF THE HEAD AND NECK***

- ***Lingual vein:***
  - the tip drain into the deep lingual vein, visible on the under surface near the midline
  - run back superficial to the hypoglossus and is joined by the ***sublingual vein*** from the gland to form ***vena comtians***
  - join the internal jugular near the greater horn of the hyoid bone



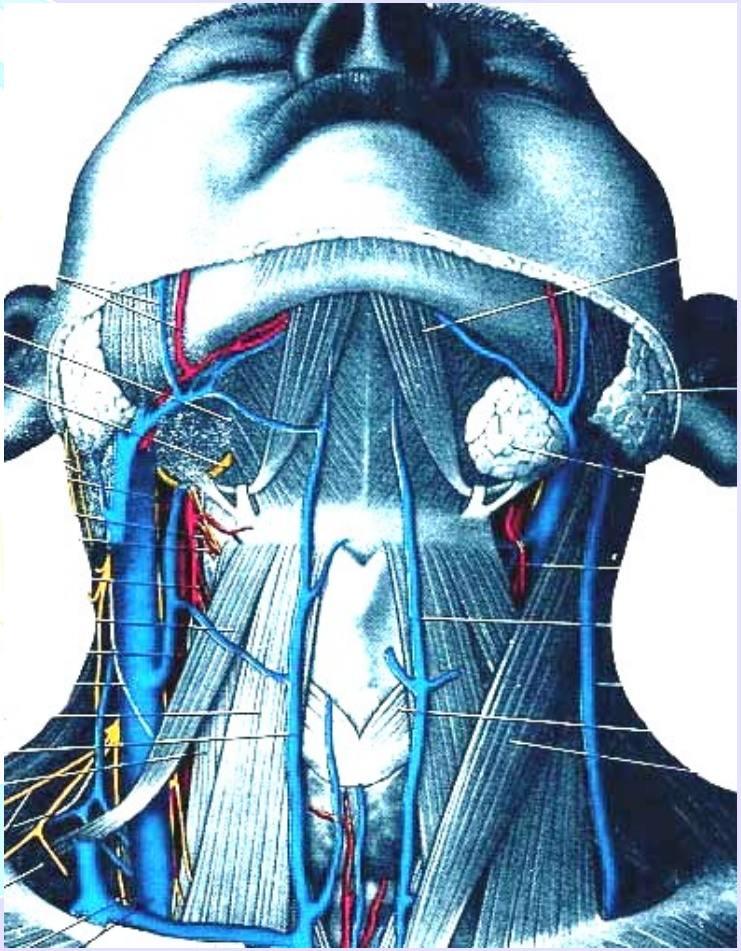
# VEINS OF THE HEAD AND NECK

- ***Superior thyroid vein:***
  - leaves the superior pole of the thyroid gland and empty in the facial or the internal jugular
- ***Middle thyroid vein:***
  - short and wide, pass from the middle of the pole directly into the internal jugular
- ***Occipital vein:***
  - join the internal jugular but more often join the vertebral or the posterior auricular



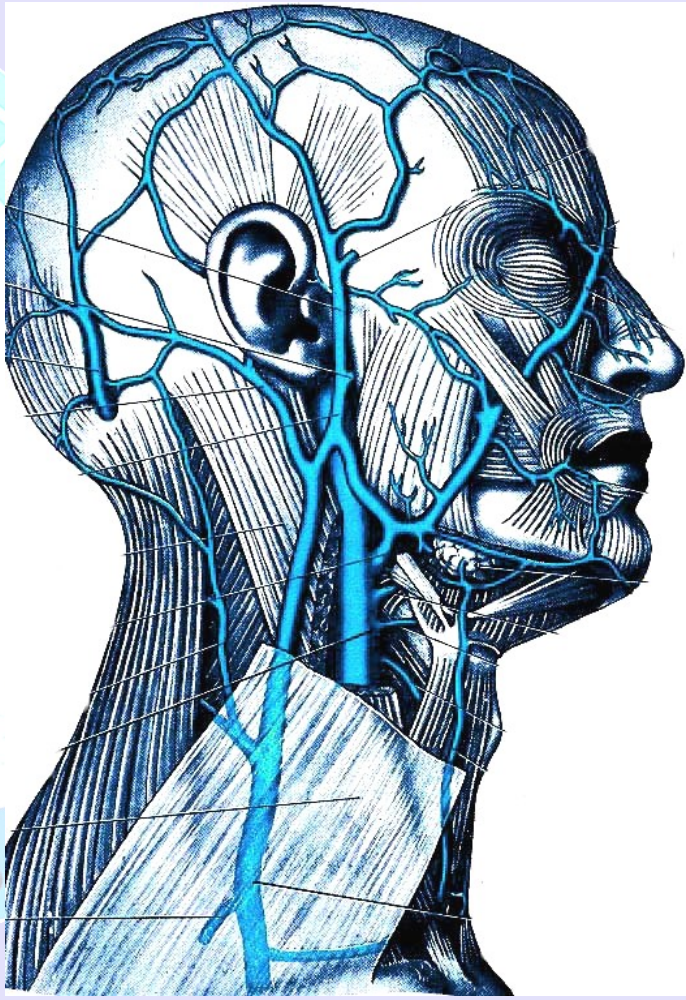
# ***VEINS OF THE HEAD AND NECK***

- ***Anterior jugular vein:***
  - start below the chin, pass beneath the platysma to the suprasternal notch.
  - Pierce the deep fascia and is connected to the other side by an anastomosing vein the jugular arch
  - angle laterally to pass deep to sternocleidomastoid and open in the external jugular vein



# ***VEINS OF THE HEAD AND NECK***

- ***External jugular vein:***
  - Begins behind the angle of the mandible by the union of the ***posterior auricular*** and ***posterior division of the retromandibular veins*****.**
  - It descend obliquely, deep to the platysma, receive the posterior external jugular vein
  - pierce the deep fascia just above the clavicle and drain into the ***subclavian vein***



# ***VEINS OF THE HEAD AND NECK***

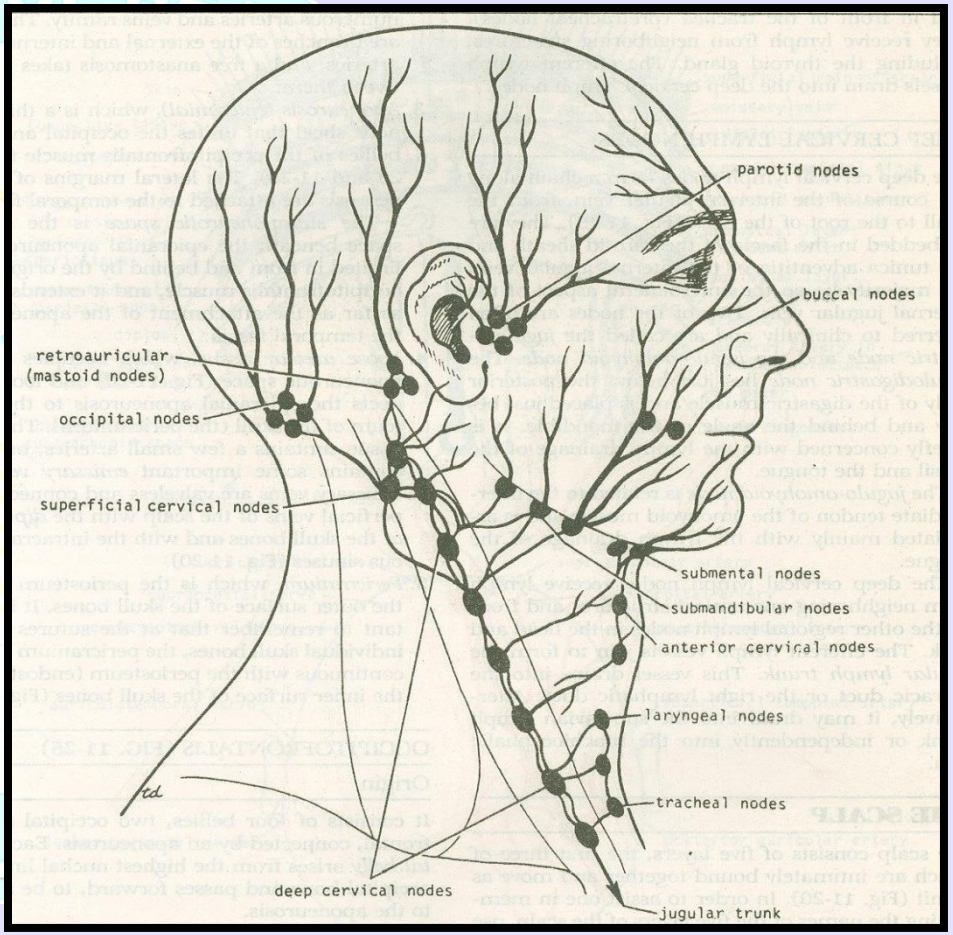
- Tributaries:
  - **Posterior auricular vein**
  - **posterior division of retromandibular vein**
  - **Posterior exernal jugular vein**
  - **transverse cervical vein**
  - **suprascapular vein**
  - **anterior jugular vein**

# LYMPH NODES OF HEAD & NECK

- Comprises of:

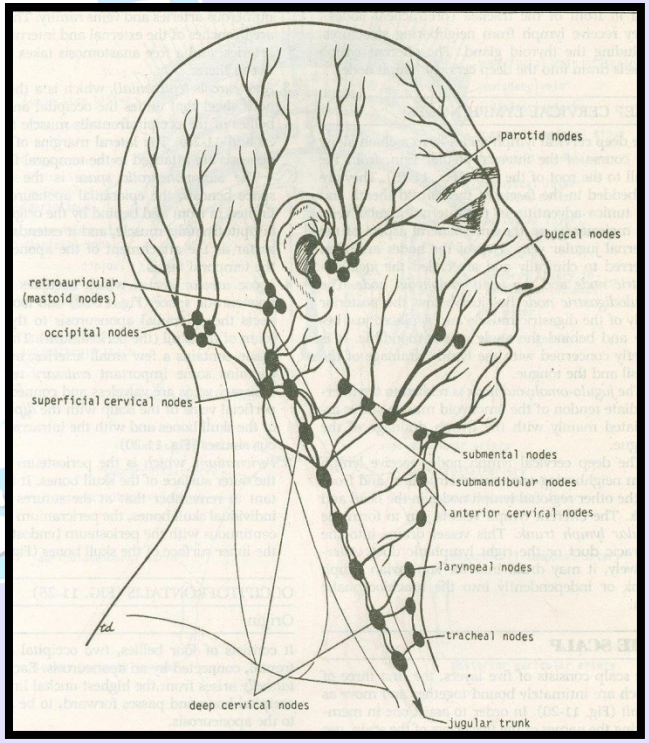
1. Regional gp.
   - The occipital
   - The retroauricular(mastoid)
   - The parotid
   - The buccal(facial)
   - The submandibular
   - The submental
   - The anterior cervical
   - The superficial cervical
   - The retropharyngeal
   - The laryngeal
   - The tracheal

- Terminal gp:
   - The deep cervical group



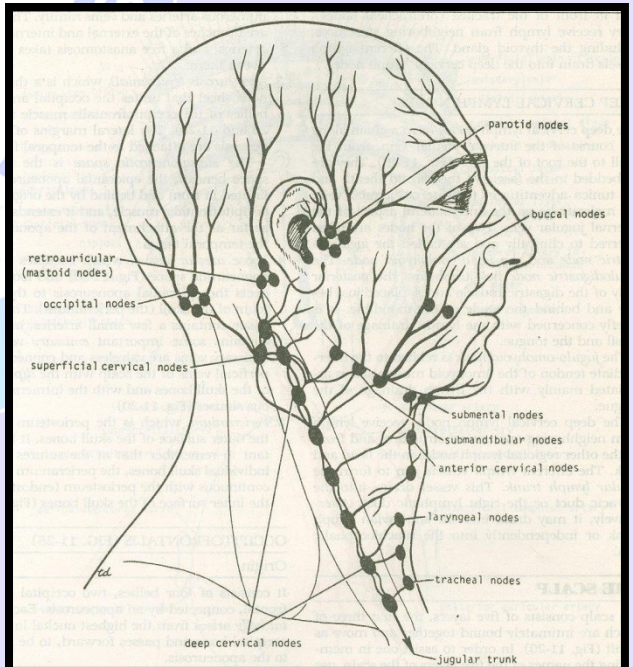
# LYMPH NODES OF HEAD & NECK

1. **The occipital:**
   - Situated over the occipital bone at the apex of posterior triangle.
   - **Afferent:** from
     - back of the scalp
   - **Efferent:** drain into
     - deep cervical lymph nodes
2. **The retroauricular(mastoid):**
   - Situated over the lateral surface of the mastoid process
   - **Afferent:** from
     - A strip of scalp above the auricle
     - Posterior wall of external auditory meatus
   - **Efferent:** drain into
     - deep cervical lymph nodes



# LYMPH NODES OF HEAD & NECK

3. **The parotid:**
   - Situated on or within the parotid gland
   - **Afferent:** from
     - A strip of scalp above the salivery gland
     - Lateral surface of the auricle
     - Anterior wall of external auditory meatus
     - Lateral parts of eyelids
   - **Efferent:** drain into
     - deep cervical lymph nodes
4. **The buccal(facial):**
   - Situated over the buccinator muscle, close to the facial vein.
   - **Efferent:** drain into
     - Submandibular lymph nodes



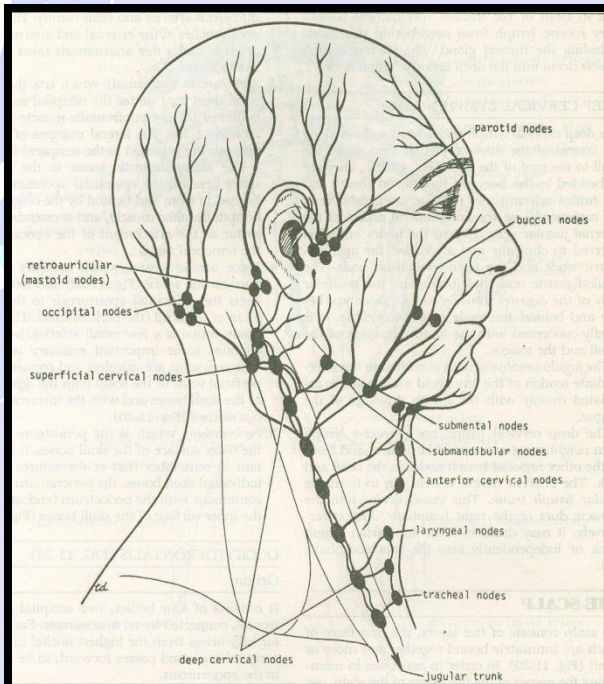
# *LYMPH NODES OF HEAD & NECK*

## 5. The submandibular:

- Situated over the superficial surface of submandibular salivary glands beneath deep fascia.

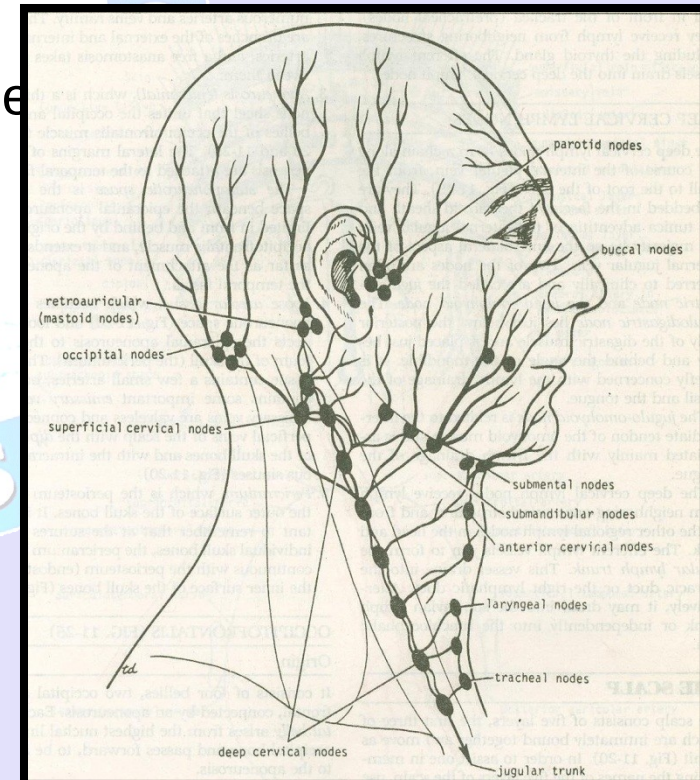
– **Afferent:** from
  - front of the scalp
  - The nose & the adjacent cheek
  - Upper & lower lips (except centre)
  - Frontal, maxillary, ethmoid air sinuses
  - Upper & lower teeth (except lower incisor)
  - The anterior 2/3 of tongue (except tip)
  - The floor of the mouth & vestibule
  - The gums

– **Efferent:** drain into
  - deep cervical lymph nodes



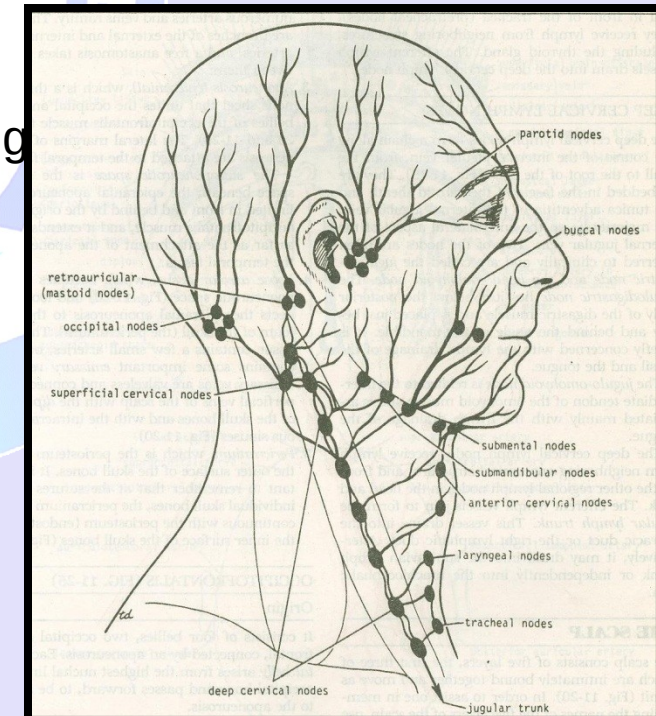
# *LYMPH NODES OF HEAD & NECK*

5. **The submental:**
   - Situated in the submental triangle b/w anterior belly of diagastric muscle.
   - **Afferent:** from
     - Tip of tongue
     - Floor of mouth beneath tip of tongue
     - The incisor
   - **Efferent:** drain into
     - Submandibular lymph nodes
     - deep cervical lymph nodes



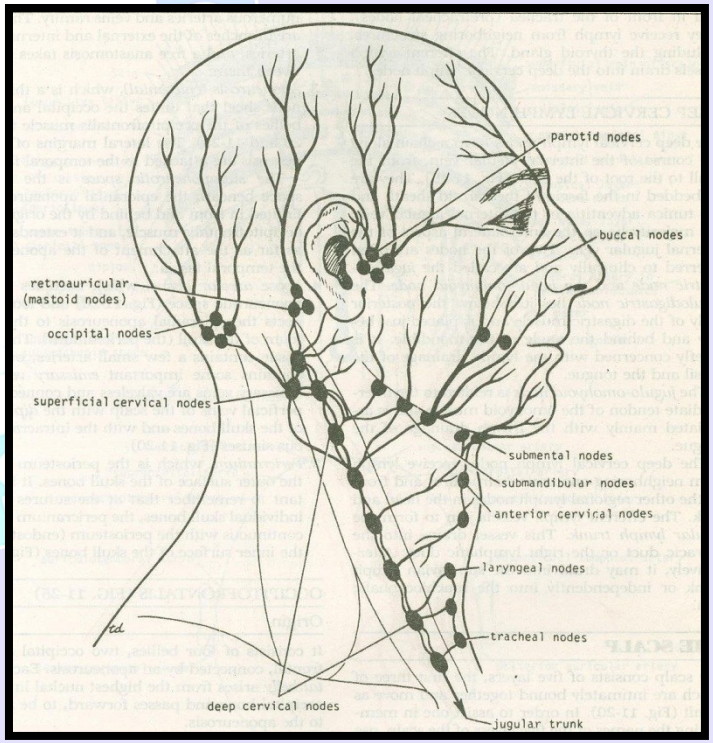
# LYMPH NODES OF HEAD & NECK

6. **The anterior cervical :**
   - Situated along the course of the anterior juglar veins.
   - **Afferent:** from
     - The skin & superficial tissues of the front of the neck
   - **Efferent:** drain into
     - deep cervical lymph nodes
7. **The superficial cervical:**
   - Situated along the course of the external jug
   - **Afferent:** from
     - The skin over the angle of jaw
     - The skin over the apex of the parotid gland
     - The lobe of the ear
   - **Efferent:** drain into
     - deep cervical lymph nodes



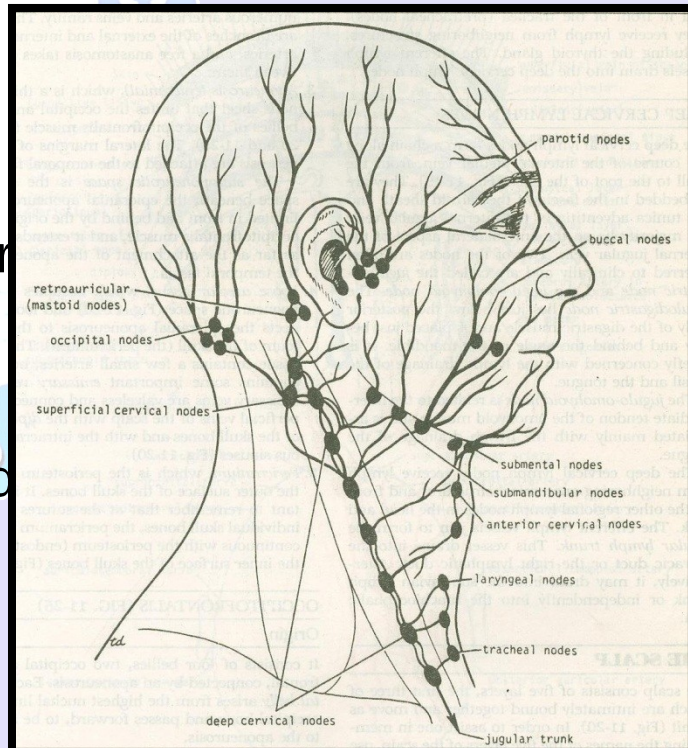
# LYMPH NODES OF HEAD & NECK

6. **The retropharyngealngeal wall & prevertebral fascia:**
   - Situated in retropharyngeal space, in the interval b/w the phar.
   - **Afferent:** from
     - Nasal part of pharynx
     - The auditory tube
     - upper part of cervical vertebrae.
   - **Efferent:** drain into
     - deep cervical lymph nodes



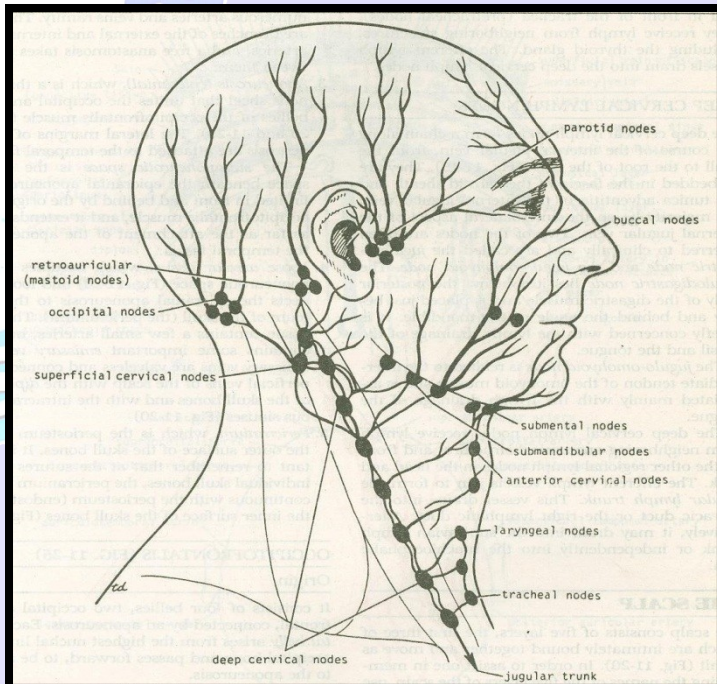
# LYMPH NODES OF HEAD & NECK

6. **The laryngeal:**
   - Situated infront of larynx on cricothyroid ligament
   - **Afferent:** from
     - Adjacent structures
   - **Efferent:** drain into
     - deep cervical lymph nodes
7. **The tracheal:**
   - Situated lateral to the trachea(paratr
   - Infront of trachea (pretracheal)
   - **Afferent:** from
     - Adjacent structures including thyroid
   - **Efferent:** drain into
     - deep cervical lymph nodes



# LYMPH NODES OF HEAD & NECK

- ***Terminal group:***
  - ***Deep cervical lymph nodes:***
    - form a chain along the course of the internal jugular vein, from skull to the root of the neck in
    - Embeded in the fascia of carotid sheath.
    - Majority lie on the anterolateral aspect of the vein.
  - **Afferent:**from
    - All regional lymph nodes
    - Adjacent anatomical structures
  - **Efferent:**
    - Efferent lymph vessels join to form jugular lymph trunk
    - Drain into thorasic duct or the right lymphatic duct
    - May drain in to brachiocephlic vein






# ***THANK YOU***